

**A GPS-BASED MATERNAL EMERGENCY RESPONSE AND
MONITORING SYSTEM FOR IMPROVING HEALTH IN UGANDA**

CASE STUDY: MUKONO DISTRICT

BY

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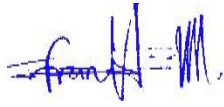
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DECLARATION

We, Muhinda Praise Francis, Nabukeera Zaharah Ramulah, and Kato Hassan, declare that the work presented in this research proposal is our original work and has not been submitted to any University or Institution of Higher Learning for any academic award. All work from other authors has been fully and properly acknowledged and cited.

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APPROVAL

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LIST OF ACRONYMS

- GIS: Geographic Information System
- WHO: World Health Organization
- UBOS: Uganda Bureau of Statistics
- IoT: Internet of Things
- PPH: Postpartum Haemorrhage
- NDP III: Third National Development Plan
- SDG: Sustainable Development Goals
- GSM: Global System for Mobile Communications
- USSD: Unstructured Supplementary Service Data
- mHealth: Mobile Health
- HIS: Health Information Systems
- SMS: Short Message Service
- VHTs: Village Health Teams
- UNICEF: United Nations Children's Fund
- GPS: Global Positioning System
- CEmONC: Comprehensive Emergency Obstetric and Newborn Care

DEFINITION OF TERMS USED

- **GPS (Global Positioning System):** A satellite-based navigation system used to provide precise, real-time geographic coordinates (latitude and longitude) to identify a mother's exact physical location during an emergency.
- **Maternal Health:** The health of women during pregnancy, childbirth, and the postpartum period. In this study, this specifically focuses on reducing mortality and addressing life-threatening complications like Postpartum Haemorrhage.
- **Emergency Response:** The coordinated process of dispatching medical assistance such as ambulances and Village Health Teams (VHTs) to a patient in immediate need of life-saving care.
- **Monitoring System:** A digital infrastructure consisting of a mobile interface and a centralized hospital dashboard used to track, observe, and manage the status and location of expectant mothers in real-time.
- **Spatial Invisibility:** A technical gap where healthcare providers lack digital visibility of a patient's location, often due to poorly mapped areas or reliance on vague verbal directions.
- **Second Delay:** The critical logistical time lost between the decision to seek care and the actual arrival of the mother at a health facility.
- **Maternal Health Informatics:** The interdisciplinary study of the design, development, adoption, and application of IT-based innovations in healthcare services for expectant mothers.
- **Real-time Analytics:** The use of data and resources as soon as they enter a system, allowing for immediate feedback and decision-making during an obstetric crisis.
- **Internet of Things (IoT):** A network of physical objects such as wearable sensors embedded with technologies for the purpose of connecting and exchanging data with other devices over the internet.

- Geospatial Integration: The process of combining GPS coordinates with existing health databases to visualize patient locations on a digital map.
- Referral Loop: A systemic failure where a patient is sent between multiple health facilities that all lack the necessary resources to treat them, caused by a lack of real-time resource visibility.

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CHAPTER ONE: INTRODUCTION

1.1 Introduction

This chapter provides the foundational framework for the proposed research, which focuses on developing a GPS-integrated maternal health system to reduce mortality rates in Mukono district, Uganda. It begins by establishing the background of the study, contextualizing the maternal health crisis and the critical role of spatial technology in modern healthcare. It then transitions to the problem statement, which critically analyses the causes and effects of spatial invisibility within current emergency maternal care workflows. Following the problem statement, the study's primary and specific objectives are defined to outline the technical milestones of the research. Furthermore, the scope is established to provide clear subject, geographical, and time boundaries for the project and finally the significance of the study is then presented through a detailed justification of its practical and theoretical contributions to the field of health informatics.

1.2 Background to the Study

The field of maternal health informatics has transitioned from static record-keeping to a dynamic integration of the Internet of Things (IoT), Global Positioning Systems (GPS), and real-time analytics. Globally, the World Health Organization (WHO) reports approximately 800 daily maternal deaths from preventable causes, with 95% occurring in low-resource settings (World Health Organization, 2023). In emerging economies like India, the deployment of GPS-tracked ambulances and IoT-enabled wearable sensors has proven effective in reducing the Second Delay, the critical interval required to reach a health facility (Thaddeus & Maine, 1994).

In Africa, digital health initiatives have primarily centred on mHealth. While programs like Kenya's Linda Mama have increased service uptake; they remain fundamentally location-blind.

In Uganda, where the maternal mortality ratio stands at 284 deaths per 100,000 live births (Uganda Bureau of Statistics, 2022), existing interventions such as UNICEF's SMS-based reminders and Kampala's eHealth referral pilots lack real-time geospatial integration. This

technical deficit prevents healthcare providers from tracking patients in labour or identifying precise coordinates during obstetric emergencies.

The current influence of GPS and IoT on Ugandan maternal health is restricted to high-level urban pilots, leaving a significant technological gap in peri-urban and rural districts. Existing systems function as static databases rather than active emergency dispatch tools, resulting in fatal navigation errors during complications like Postpartum Haemorrhage (PPH).

Mukono District, the case study for this research, epitomizes these challenges. As a region characterized by poorly mapped rural villages in Nama and isolated island communities in Koome, verbal directions are frequently unreliable. Consequently, maternal referrals to Mukono General Hospital suffer from spatial invisibility. This study will aim to develop a GPS-enabled solution within Mukono to provide a scalable, data-driven model for reducing maternal mortality through precise, real-time location tracking.

1.3 Statement of the Problem

Despite national interventions, Uganda's maternal mortality ratio remains critically high at 284 deaths per 100,000 live births (Uganda Bureau of Statistics, 2022). In Mukono District, a primary driver of these fatalities is the Second Delay, the time lost in locating and transporting a mother during an obstetric emergency. Current response workflows are location-blind, relying on manual records and vague verbal directions, which prevents the timely dispatch of ambulances and Village Health Teams (VHTs). This lack of spatial visibility frequently turns manageable complications, such as Postpartum Haemorrhage, into avoidable tragedies.

The root cause of this crisis is spatial information invisibility. The absence of a digital infrastructure to transmit precise GPS coordinates leads to fatal navigation errors in poorly mapped areas like Nama and isolated island communities like Koome. This is exacerbated by fragmented communication and the reliance on physical paper cards, which restricts access to vital clinical history during transit. Consequently, mothers lose the critical therapeutic window for intervention, often falling into inefficient referral loops between facilities that lack necessary resources. This systemic failure results in unassisted births and

high risks of sepsis, primarily affecting rural mothers in Mukono and hindering the achievement of Sustainable Development Goal 3 (United Nations, 2015).

To address these challenges, this research will propose a GPS-Enabled Maternal Health Information System for Mukono District. By integrating a mobile interface for mothers with a web-based hospital dashboard, the system will utilize GIS technology to provide real-time tracking and automated resource identification. This solution ensures a mother's precise location is visible to medical responders before she arrives at the facility, effectively bridging the spatial gap in emergency maternal care.

Despite the proliferation of mobile health (mHealth) initiatives in Uganda, such as UNICEF's mTrac and various SMS-based antenatal reminder systems, a significant technological void remains in the area of real-time emergency navigation. Existing systems are fundamentally "location-blind," functioning primarily as static databases for record-keeping rather than active tools for emergency dispatch.

Current literature focuses heavily on the clinical causes of maternal mortality, yet there is a lack of integrated research into how Geospatial Informatics can specifically resolve the Second Delay, the logistical time lost while health workers and ambulance drivers struggle to locate a mother in distress due to vague verbal directions or poorly mapped terrains. In Mukono District, the geographical isolation of island communities and rural sub-counties like Nama exacerbates this "spatial invisibility". This study will, therefore, address this gap by designing a system that synchronizes a mother's real-time GPS coordinates with a centralized hospital dashboard, providing a first-of-its-kind template for event-driven spatial health informatics in Uganda's public health sector.

1.4 Objectives of the Study

1.4.1 General Objective

The main objective of this research is to design and develop a GPS-Enabled Maternal Health Information System for Mukono District to eliminate spatial invisibility and reduce the Second Delay during obstetric emergencies.

1.4.2 Specific Objectives

- (i) (i) To study and analyse the strengths and weaknesses of the current maternal emergency response and referral workflows within Mukono General Hospital and its surrounding health centres.
- (ii) (ii) To determine and specify the functional and technical requirements for a real-time geospatial tracking and alert system tailored for the terrain of Mukono District.
- (iii)(iii) To design a GPS-integrated maternal health system consisting of a mobile emergency interface for mothers and a centralized navigation dashboard for hospital administrators.
- (iv)(iv) To implement, test, and validate the new system through simulated emergency scenarios to measure its effectiveness in improving response times and coordination.

1.5 Research Questions

To achieve the objectives of this research, the following questions will be addressed;

- (i) What are the current strengths and weaknesses of the existing maternal emergency response and referral workflows in Mukono General Hospital and its surrounding health canters?
- (ii) What are the specific functional and technical requirements necessary to develop a real time geospatial tracking alert system that accounts for the unique terrain of Mukono District?
- (iii) How can GPS-integrated system be designed to provide an intuitive mobile interface for expectant mothers and a comprehensive navigation?
- (iv) How efficient is the implemented system in reducing emergency response times and improving coordination when tested through simulated maternal emergency scenarios?

1.6 Scope of the Study

1.6.1 Subject Scope

The subject scope of this research focuses on the design and development of a GPS-Enabled Maternal Health Information System. The system is to be developed to address the spatial

invisibility of expectant mothers during obstetric emergencies, specifically targeting the reduction of the Second Delay that is time taken to reach a health facility. The primary stakeholders of the proposed system include:

- **Expectant Mothers:** The end-users who require emergency tracking and rapid response.
- **Healthcare Providers:** Midwives and administrators at Mukono General Hospital who will monitor the real-time navigation dashboard.
- **Emergency Responders:** Ambulance drivers and Village Health Teams (VHTs) who will utilize the GPS data for precise navigation.
- **District Health Officers:** Who will use the generated geospatial data for better resource allocation and health planning.

1.6.2 Geographical Scope

The research will be conducted within Mukono District, Central Uganda. This case study was chosen due to its unique geographical diversity, featuring rapidly urbanizing centres alongside poorly mapped rural sub-counties like Nama and isolated island communities in Koome on Lake Victoria. Mukono exemplifies the referral gap where high maternal populations face significant transit hurdles to reach Mukono General Hospital. The district's varied terrain provides a rigorous environment to test the accuracy and reliability of GPS and GIS technologies in low-resource, high-stakes settings.

1.6.3 Time Scope

This research is scheduled to be conducted over a period of 12 weeks, structured into three intensive technical phases:

Weeks 1–4: Research and Requirements. This phase involves the literature review, gathering functional requirements from stakeholders in Mukono, and finalizing the system architecture.

Weeks 5–8: System Development. This phase focuses on the core technical implementation, including geospatial database design, mobile interface coding, and the integration of GPS and Mapping modules.

Weeks 9-12: Validation and Documentation. The final phase is dedicated to system testing, user validation within Mukono District, and the completion of the final project submission.

1.7 Significance of the Study

The urgent necessity of this study is underscored by the reality that maternal mortality in Uganda is not merely a medical failure, but a logistical and information failure. While medical protocols for treating complications exist, the inability to locate and reach a mother in distress remains a critical bottleneck. This research is essential because it moves maternal health from a reactive paper-based model to a proactive spatial-awareness model.

Current maternal health interventions in Uganda primarily rely on mHealth (Mobile Health) platforms that focus on one-way communication, such as SMS reminders for antenatal visits. However, a significant technical gap exists, these systems are blind to the mother's physical location during a crisis.

Most existing studies focus on clinical causes of death but overlook the Geospatial Informatics required to solve the Second Delay which is reaching the facility (Faujdar & Kumar, 2020). There is currently no integrated platform in Uganda's public hospitals that links a mother's real-time GPS coordinates to a centralized hospital dashboard for immediate emergency dispatch.

The implementation of this system offers immediate, tangible improvements to the maternal healthcare process:

- **Precision Emergency Response:** By replacing vague verbal directions with precise GPS coordinates, the system minimizes the time ambulances or Village Health Teams (VHTs) spend searching for a patient's home.
- **Intelligent Referral Mapping:** The system can identify the nearest facility with available beds and the correct blood type, preventing the common referral merry-go-round where mothers are bounced between full hospitals.
- **Data-Driven Policy:** Aggregated GPS data will allow the Ministry of Health to identify mortality hotspots, enabling them to place new clinics or ambulances exactly where they are needed most.

This study contributes to the academic body of knowledge in the following ways:

- **GIS-Healthcare Integration:** It provides a new framework for applying Geographic Information Systems to maternal health in resource-constrained environments.
- **Low-Bandwidth Optimization:** Academically, this research explores the synchronization of spatial data over GSM/USSD networks, providing a model for how complex GPS data can be transmitted in areas with poor internet connectivity.
- **Informatics Evolution:** It advances the study of Health Information Systems (HIS) by shifting the focus from static data storage to real-time, event-driven architecture.

This research serves as a pioneer for Spatial Health Informatics in East Africa. By successfully integrating GPS tracking into public hospital workflows, this project provides a scalable template that can be adapted for other emergencies, such as accident response or epidemic tracking. It directly supports the Uganda National Development Plan (NDP III) (National Planning Authority, 2020), proving that homegrown, location-aware technology is the most effective tool for reducing the national maternal mortality ratio.

1.8 Chapter Summary

Chapter One has established the foundational framework for this research proposal by identifying the urgent need for a spatial intervention in Uganda's maternal healthcare. The Problem Statement highlighted the fatal consequences of delayed emergency response and the inefficiency of current paper-based or non-localized systems. The Objectives of the study were clearly defined, focusing on the design, development, and testing of a GPS-tracking module and a centralized maternal database.

Furthermore, the Scope and Justification emphasized the technical feasibility and the potential socio-economic impact of the system in alignment with the Uganda National Development Plan (NDP III) (National Planning Authority, 2020) and Sustainable Development Goal 3. Ultimately, this chapter sets the stage for the subsequent review of existing literature and the detailed methodology that will be employed to bring this life-saving technical solution to fruition.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter presents a critical review of literature related to the design and development of a GPS-based maternal emergency response and monitoring system, with specific focus on Mukono District, Uganda. The review examines theoretical and empirical studies in maternal health informatics, situates maternal mortality within both global and national contexts, and analyses the role of geospatial technologies in emergency obstetric care.

Particular emphasis is placed on the interaction between geography, infrastructure, and health system responsiveness in Mukono District. The district's rural sub-counties, island communities, and referral pathways to Mukono General Hospital are examined as key determinants of emergency response efficiency. The chapter concludes by synthesizing the literature to identify gaps that justify the development of a real-time GPS-enabled maternal emergency system.

2.2 Literature Themes

The literature reviewed for this study is organized into four thematic areas that represent the intersection of technology and maternal healthcare. These themes provide a structured analysis of the current state of research and help contextualize the proposed GPS-based intervention within the broader field of health informatics:

- **Evolution of Maternal Health Informatics:** This theme explores the transition from paper-based record-keeping to digital health systems, specifically focusing on how Mobile Health (mHealth) has been utilized in low-resource settings to improve antenatal care.
- **The Three Delays Model and Spatial Barriers:** This theme examines the sociological and logistical frameworks that explain maternal mortality, with a specific focus on the Second Delay, the time lost in transit, and the role of geographical factors in healthcare access.
- **Applications of Geospatial Technology in Emergency Care:** This theme reviews global and regional empirical evidence on the effectiveness of Global Positioning

Systems (GPS) and Geographic Information Systems (GIS) in reducing ambulance response times and improving patient tracking.

- **The Technical Landscape of Health Information Systems in Uganda:** This theme analyses the strengths and limitations of existing national and district-level systems in Uganda, highlighting the current lack of real-time geospatial coordination in local health facilities.

2.3 Conceptual Framework: Maternal Health Informatics & Three Delays

2.3.1 Evolution of Health Information Systems in Maternal Care

Health Information Systems (HIS) have evolved from static administrative tools into dynamic platforms that support real-time decision-making. Early systems primarily focused on patient registration and aggregate reporting, limiting their utility in emergency scenarios. In Uganda, the Health Management Information System (HMIS) reflects this limitation, as it largely supports retrospective data analysis rather than real-time intervention.

The emergence of mobile health (mHealth) technologies has significantly improved communication in maternal healthcare. Initiatives such as UNICEF's mTrac and MamaUgandi have enhanced antenatal care attendance and drug monitoring through SMS-based platforms. However, these systems remain fundamentally location-blind, lacking the ability to provide spatial intelligence during emergencies.

Recent studies highlight that while mHealth improves information dissemination, it does not address geospatial coordination, which is critical in emergency response. In Mukono District, Village Health Teams (VHTs) rely heavily on mobile phones, yet the absence of location-aware technologies limits their effectiveness in guiding emergency responders.

This indicates a clear technological gap: the transition from communication-based systems to location-aware, real-time emergency systems remains largely unaddressed.

2.3.2 Theoretical Foundation: The Three Delays Model

The Three Delays Model provides a foundational framework for understanding maternal mortality.

It identifies three critical barriers:

- **The First Delay:** The time taken by the mother or her family to recognize a complication and decide to seek professional care.
- **The Second Delay:** The logistical interval required to reach an appropriate health facility after the decision to seek care has been made.
- **The Third Delay:** The time elapsed between arrival at the facility and the receipt of adequate, life-saving medical intervention.

While all three delays are significant, this study focuses on the second delay, which is heavily influenced by geographical and infrastructural factors.

In the context of Mukono District, the second delay is compounded by what this study conceptualizes as spatial invisibility the inability of healthcare providers to accurately locate patients during emergencies. This transforms distance from a physical barrier into an information failure.

Empirical evidence from Uganda shows that delays in reaching facilities are a major contributor to maternal mortality, particularly in rural and peri-urban settings. In Mukono, poorly mapped settlements such as Nama and isolated island communities like Koome intensify this challenge.

Thus, addressing the second delay requires not only transport solutions but also geospatial intelligence systems.

2.3.3 Addressing the Second Delay through Geospatial Informatics

The focus of this research is specifically centred on mitigating the Second Delay. In the context of Mukono, this delay is not merely a product of physical distance, but of **Spatial Invisibility**, a technical gap where healthcare providers cannot digitally visualize or navigate to a patient's precise location.

By applying Geospatial Integration, the framework transforms distance from a physical barrier into a manageable data point. The proposed system replaces vague verbal directions and manual records with precise GPS coordinates, thereby optimizing the "Referral Loop" and ensuring that mothers reach Comprehensive Emergency Obstetric and Newborn Care (CEmONC) within the critical therapeutic window.

2.3.4 Variables of the Proposed System

The framework operates through three distinct levels of interaction:

- Independent Variable (The Technical Intervention): A GPS-integrated system featuring a mobile emergency interface for mothers and a centralized web-based dashboard for hospitals.
- Moderating Variables (Environmental Constraints): Local factors such as the unique terrain of Mukono (e.g., the Koome island communities), GSM/USSD network reliability, and the operational readiness of Village Health Teams (VHTs).
- Dependent Variable (The Health Outcome): A measurable reduction in emergency response times, the elimination of navigation errors, and an overall decrease in the maternal mortality ratio within the district

2.4 Maternal Health Context in Mukono District

2.4.1 Geographical and Demographic Characteristics

Mukono District is located in Central Uganda and exhibits significant geographical diversity, including urban centres, rural settlements, and island communities on Lake Victoria. This diversity creates uneven access to healthcare services.

With an estimated population of approximately 700,000, the district has a high proportion of women of reproductive age, increasing demand for maternal health services.

However, geographical constraints particularly in island communities introduce systemic delays in accessing care. Transport limitations, long distances, and unreliable

communication networks collectively increase the risk of maternal complications and mortality.

2.4.2 Health Facilities and Referral System

Mukono General Hospital serves as the primary referral facility for comprehensive emergency obstetric and newborn care (CEmONC). Lower-level facilities provide basic services but often lack critical resources such as surgical capacity and blood transfusion services.

The referral system is largely manual and fragmented, relying on phone communication and informal directions. This results in inefficiencies, particularly during emergencies.

Studies indicate that ambulance drivers may spend up to 60 minutes locating patients due to unclear directions. This delay highlights a key weakness: the absence of structured, location-based referral systems.

2.4.3 Village Health Teams (VHTs) and Their Role

Village Health Teams play a crucial role in bridging communities and formal healthcare systems. They facilitate referrals, monitor pregnancies, and provide health education.

Despite their importance, VHTs face significant operational challenges, including lack of digital tools and limited logistical support. Their reliance on verbal communication further contributes to spatial inefficiencies.

Integrating VHTs into a GPS-enabled system would significantly enhance their ability to coordinate emergency responses and improve maternal outcomes.

2.5 Review of GPS and GIS Applications in Maternal Health

2.5.1 Global Evidence on Geospatial Technologies in Emergency Care

Geospatial technologies such as GIS and GPS have transformed emergency response systems globally. They enable real-time tracking, route optimization, and spatial analysis of healthcare access.

In high-income countries, GPS-enabled ambulance systems have significantly reduced response times. Additionally, GIS has been used to identify underserved populations and optimize healthcare resource allocation.

These applications demonstrate that location intelligence is a critical component of effective healthcare delivery.

2.5.2 GPS-Enabled Maternal Health Initiatives in Low-Resource Settings

In sub-Saharan Africa, several pilot projects have explored GPS-based maternal health solutions.

In Tanzania, GPS integration reduced response times by approximately 18 minutes. In Nigeria, the MNETI initiative achieved a significant reduction in transport time. In Kenya, GPS systems improved referral coordination despite infrastructural challenges.

These studies demonstrate the feasibility and effectiveness of GPS technologies, even in low resource environments. However, most projects remain pilot-based and lack long-term scalability.

2.5.3 Challenges of GPS Integration in Rural Uganda

Despite its potential, GPS implementation in Mukono faces several constraints:

- Limited network coverage, especially in island areas
- Low smartphone penetration
- Data privacy and security concerns
- Dependence on donor-funded pilot projects

These challenges highlight the need for a context-aware system design, incorporating offline functionality, low-bandwidth optimization (e.g., USSD/SMS), and sustainable deployment models.

2.6 Existing Maternal Health Information Systems in Uganda & Mukono

2.6.1 National Systems

Uganda's HMIS and eLMIS systems support health data management and logistics. However, they are not designed for real-time emergency coordination.

2.6.2 District-Level Practices

At the district level, communication is largely informal, often relying on phone calls and messaging platforms such as WhatsApp. These methods lack structure, data integration, and spatial capabilities.

Currently, no system in Mukono District supports:

- Real-time patient tracking
- Automated ambulance dispatch
- Geospatial decision-making

2.6.3 Role of NGOs and Private Sector

NGOs support maternal health through transport schemes and awareness programs. However, these initiatives are fragmented and not integrated into a unified system.

A centralized GPS-based platform could enhance coordination and improve efficiency across stakeholders.

2.7 Literature Gap: Spatial Invisibility

2.7.1 Spatial Invisibility in Mukono: A Critical Gap

Spatial invisibility emerges as a central challenge in maternal healthcare delivery in Mukono District. It refers to the inability of healthcare providers to accurately identify and navigate to a patient's location.

This issue is driven by:

- Lack of formal addressing systems
- Poorly mapped rural areas
- Geographical isolation of island communities
- Absence of real-time location tracking systems

Studies show that significant time is lost locating patients before transport begins. In emergency conditions such as postpartum haemorrhage, these delays can be fatal.

Thus, spatial invisibility represents a critical intersection between geography, technology, and healthcare failure.

2.8 Synthesis and Research Gap

The reviewed literature reveals several key insights:

- The second delay is a major contributor to maternal mortality in Uganda
- Existing mHealth systems lack geospatial capabilities
- GPS technologies are effective but underutilized in low-resource settings
- Mukono District presents unique geographical challenges

Despite these findings, there is no integrated, GPS-based maternal emergency response system tailored to Mukono District.

This study addresses this gap by proposing a system that:

- Integrates mobile and web platforms and enables real-time location tracking.
- Supports coordinated emergency response as well as optimization for low-resource environments.

2.9 Chapter Summary

This chapter has critically reviewed literature on maternal health informatics, the Three Delays Model, and geospatial technologies in healthcare. It has highlighted the limitations of existing systems and the unique challenges faced in Mukono District.

The concept of spatial invisibility has been identified as a key barrier to effective emergency response. The absence of real-time, location-aware systems represents a significant research and practical gap.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the methodology used in the study titled “A GPS-Based Maternal Emergency Response and Monitoring System for Improving Health in Uganda (Case Study: Mukono District)”. The chapter describes the research design, study area, study population, sample size determination, sampling techniques and procedures, data collection methods and instruments, validity and reliability of instruments, data quality control measures, data analysis techniques, ethical considerations, and limitations of the study.

This structure ensures that the study produces valid, reliable, and replicable findings aligned with the objectives of analysing current maternal emergency systems, designing a GPS-based solution, and evaluating its effectiveness.

3.2 Research Design

The study will adopt a mixed-methods approach combining a descriptive cross-sectional research design with a Prototyping System Development Model. The descriptive phase will involve collecting data at a single point in time to assess the current status and inefficiencies of maternal emergency response systems.

The integration of a prototyping model is essential for this IT-based intervention, as it will allow for iterative testing and user feedback during the design of the GPS interface. Prototyping is particularly effective in complex system designs where user requirements, such as real-time navigation, may evolve during the development process (Kendall, 2018). This combination will ensure that the findings from the descriptive phase form a rigorous foundation for the implementation and testing of the GPS-based system.

3.3 Study Area and Target Population

3.3.1 Study Area

The study will be conducted in Mukono District, Uganda, which is located in the central region of the country. Mukono District is characterized by a combination of

urban, peri-urban, and rural settings, including hard-to-reach areas with limited infrastructure and transportation networks.

These geographical constraints introduce systemic delays and "spatial invisibility," where healthcare providers cannot accurately locate patients. Geographical accessibility remains a critical predictor of maternal outcomes in rural Uganda, necessitating the use of location-aware digital health interventions (Makanga et al., 2016).

The district has a structured healthcare system comprising both public and private health facilities. Specifically, it includes Mukono General Hospital, which serves as the major government referral facility, as well as approximately twelve registered private hospitals and clinics providing various levels of healthcare services. These facilities offer essential maternal health services such as antenatal care, delivery services, and emergency obstetric care.

Mukono District is selected as the study area due to its relevance to the research problem. The district experiences notable maternal health challenges, including delays in accessing emergency care, which are often exacerbated by geographical barriers such as poor road networks and dispersed settlements. These conditions contribute to inefficiencies in emergency response systems, particularly during obstetric emergencies.

Furthermore, Mukono represents a typical Ugandan district in terms of healthcare structure and service delivery, making it suitable for generalizing findings to similar settings. The presence of both government and private hospitals provides a comprehensive context for analysing variations in emergency response systems and identifying areas for improvement.

Mukono District therefore represents a typical Ugandan district in terms of healthcare structure, making it suitable for generalizing findings to similar settings.

3.3.2 Target Population

In this study, the population consists of all general government hospitals and known private hospitals in Mukono District, Uganda.

Within these health facilities, the study will focus on key categories of respondents who are directly involved in maternal healthcare delivery and emergency response systems. These include hospital administrators, midwives and nurses, ambulance drivers and emergency responders, as well as expectant mothers. These groups are selected because they actively participate in maternal emergency processes and can provide reliable information regarding system performance, response challenges, and coordination gaps.

To ensure that only relevant and functional facilities are included in the study, the following inclusion criteria will be applied:

- ✓ The hospital must have been operational for at least one year to ensure consistency in service delivery.
- ✓ The facility must be licensed by the Ministry of Health to guarantee compliance with national healthcare standards.
- ✓ The hospital must provide maternal health services such as antenatal care, delivery services, and emergency obstetric care.
- ✓ The facility must have a minimum capacity of 20 beds to ensure adequate patient flow and operational relevance.
- ✓ The estimated number of eligible hospitals in Mukono District ranges between 10 and 15 facilities, based on district health structures and administrative records.

As part of the study implementation, data will be collected from selected hospitals within this population, including Mukono Church of Uganda Hospital, C-Care (IMC) Hospital, AAR Pearl Hospital, and Mukono General Hospital (government facility). These facilities are considered because they represent both public and private healthcare providers and are actively involved in maternal healthcare service delivery within the district.

This study population is appropriate because it captures all key stakeholders involved in maternal emergency response systems, while the selected hospitals provide practical settings for obtaining in-depth, real-world data. This ensures that the findings reflect both system-wide patterns and facility-level operational realities.

3.3.3 Sample Size and Sampling Strategy

3.3.3.1 Sample Size Determination

A census approach will be adopted, whereby all identified government and private hospitals in Mukono District will be included.

The use of a census approach is justified on several grounds. First, the total number of hospitals within the study area is relatively small and manageable, making it feasible to include all eligible facilities. Second, the census approach ensures full representation of the study population, thereby enhancing the reliability and generalizability of the findings. Lastly, it eliminates sampling bias, as no subset of the population is excluded from the study.

Data collection will be conducted from accessible hospitals within the district, including Mukono Church of Uganda Hospital, C-Care (IMC) Hospital, AAR Pearl Hospital, and Mukono General Hospital. These facilities form part of the overall population and provide practical entry points for obtaining field data.

3.3.3.2 Sampling Techniques and Procedures

The study will employ a census sampling technique at the institutional level, whereby all identified public and private hospitals in Mukono District will be included in the study.

The use of a census approach at the hospital level is consistent with the study's objective of obtaining comprehensive information on maternal emergency response systems across the district. Given the relatively small number of eligible hospitals, this approach ensures full coverage and minimizes the risk of excluding relevant facilities.

However, within each hospital, different sampling techniques will be applied to select respondents based on their roles and relevance to the study:

Hospital administrators and healthcare workers, including midwives and nurses, will be selected using purposive sampling. This technique is appropriate because these respondents possess specialized knowledge and direct experience with maternal emergency response processes, making them key informants for the study.

On the other hand, expectant mothers receiving services at the selected hospitals will be selected using simple random sampling. This approach ensures that each eligible patient has an equal chance of being included in the study, thereby reducing selection bias and enhancing the representativeness of patient perspectives.

The data will be collected in selected accessible hospitals within the study area, including Mukono Church of Uganda Hospital, C-Care (IMC) Hospital, AAR Pearl Hospital, and Mukono General Hospital. These facilities provide a balanced representation of both public and private healthcare providers in Mukono District.

The sampling procedure will involve several steps. First, a comprehensive list of hospitals will be obtained from the Mukono District Health Office. Second, the hospitals will be categorized into public and private facilities to ensure proper representation. Third, official permission will be sought from hospital administrators before conducting the study. Fourth, eligible respondents within each hospital will be identified based on their roles and involvement in maternal healthcare services. Finally, data collection instruments, including questionnaires and interview guides, will be administered to the selected respondents.

This sampling approach will ensure that the study captures both institutional-level and individual-level perspectives, thereby enhancing the validity, reliability, and relevance of the findings.

3.4 Data Collection Instruments & Methods

3.4.1 Data Collection Methods

Here, a combination of quantitative and qualitative data collection methods will be employed to obtain comprehensive information on maternal emergency response systems and the effectiveness of the proposed GPS-based solution.

The study will utilize four main data collection methods, that is; questionnaires, interviews, observation, and system testing. The use of multiple methods allows for triangulation, thereby improving the validity and reliability of the findings.

Structured questionnaires will be used to collect quantitative data from healthcare workers and expectant mothers in the selected hospitals, including Mukono Church of Uganda Hospital, C-Care (IMC) Hospital, AAR Pearl Hospital, and Mukono General Hospital. The

questionnaires will capture measurable variables such as emergency response times, system efficiency, and availability of emergency tools and resources. This approach is appropriate because it will enable the collection of standardized data from a relatively large number of respondents, making it suitable for statistical analysis and comparison across facilities. The questionnaire method directly supports Objective One and Two of the study.

In addition, semi-structured interviews will be conducted with hospital administrators and emergency responders to collect qualitative data. These interviews will focus on exploring workflow challenges, system limitations, coordination gaps, and decision-making processes during maternal emergencies. The interview method is particularly useful in this study because it allows for in-depth exploration of complex issues that cannot be fully captured through structured questionnaires. It will therefore complement the quantitative data by providing detailed contextual insights aligned with Objective One Two.

Observation will also be employed as a data collection method to assess real-time practices within hospital settings. We, the researchers, will observe current emergency response processes, communication systems, and coordination mechanisms used during maternal emergencies. This will provide firsthand evidence of how systems operate in practice, rather than relying solely on self-reported data. This method will support Objective One by validating the actual workflow processes within the hospitals.

Furthermore, system testing will be conducted as part of the study to evaluate the performance of the proposed GPS-based maternal emergency response system. This will involve simulated emergency scenarios to measure system responsiveness, accuracy of location tracking, and improvements in response time. Field testing is vital because signal attenuation in rural or poorly mapped areas can significantly impact the reliability of emergency dispatch coordinates (Zickuhr, 2013).

The combination of these data collection methods will therefore ensure a comprehensive analysis of both existing maternal emergency response systems and the proposed technological solution, thereby strengthening the credibility and validity of the study findings.

3.4.2 Data Collection Instruments

The main instruments that will be used in the study include structured questionnaires, interview guides, and observation checklists. These instruments are designed to collect both quantitative and qualitative data necessary for analysing maternal emergency response systems and evaluating the proposed GPS-based solution.

The primary instrument for collecting quantitative data will be a structured questionnaire. The questionnaire will be administered to healthcare workers and expectant mothers in selected hospitals within Mukono District. It will be divided into several sections to ensure comprehensive data collection. The first section will capture demographic information of respondents, such as age, role, and level of experience. The second section will focus on existing maternal emergency response systems within the hospitals, including current practices and procedures. The third section will examine challenges faced in handling maternal emergencies, such as delays in response, communication barriers, and resource limitations. The final section will gather information on system requirements and expectations for improving emergency response through the proposed GPS-based system. The structured nature of the questionnaire will ensure consistency in responses and facilitates quantitative analysis aligned with the study objectives.

In addition to questionnaires, an interview guide will be used to collect qualitative data from hospital administrators and emergency responders. The interview guide will consist of open-ended questions designed to allow respondents to freely express their experiences and perspectives. The questions will focus on key areas such as challenges in emergency response, system inefficiencies, coordination issues, and gaps in existing maternal healthcare systems. This will enable us the researchers to obtain in-depth insights and detailed explanations that cannot be captured through closed-ended questions.

Furthermore, an observation checklist will be used to systematically record real-time practices within hospital settings. The checklist will capture specific elements such as emergency response steps followed during maternal emergencies and the communication flow between healthcare providers. This will provide objective evidence of workflow since it will allow us to directly observe how emergency systems operate in practice, thereby validating information obtained from questionnaires and interviews.

The application of these instruments will ensure comprehensive data collection by combining structured measurement with in-depth qualitative insights. This will enhance the reliability and validity of the study findings while ensuring alignment with the research objectives.

3.5 Reliability and Validity of Instruments

3.5.1 Validity of Instruments

Ensuring validity is essential in research because it determines the credibility and accuracy of the data collected. In this study, validity will be carefully addressed to ensure that the instruments used effectively capture information related to maternal emergency response systems and the proposed GPS-based solution.

Two main types of validity will be considered in this study, namely content validity and construct validity.

Content validity will be ensured through expert review. The data collection instruments, including questionnaires, interview guides, and observation checklists, will be examined by specialists in information technology and healthcare, particularly those with experience in maternal health systems. These experts will assess whether the items included in the instruments are relevant, clear, and comprehensive in addressing maternal emergency response processes, system challenges, and technological requirements.

Construct validity, on the other hand, will be ensured by aligning all instrument items with the study objectives. Each section of the questionnaire, interview guide, and observation checklist will be designed to correspond directly to specific objectives, such as assessing existing emergency response systems, identifying system challenges, and evaluating requirements for the GPS-based system. This alignment will thereby ensure that the data collected is relevant and meaningful for addressing the research problem.

By ensuring both content and construct validity, the study will enhance the accuracy and credibility of the data collected, thereby strengthening the overall quality of the research findings.

3.5.2 Reliability of Instruments

This will ensure that the findings of the study are dependable and not influenced by random errors or inconsistencies in measurement.

In this study, reliability will be ensured through pilot testing and statistical analysis of the data collection instruments. A pilot study will be conducted in any other two hospitals located in Mukono District but with similar characteristics to those being studied. The purpose of the pilot test will be to assess the clarity, consistency, and usability of the questionnaires, interview guides, and observation checklists. It will also help identify any ambiguities or difficulties that respondents may encounter while answering the questions.

Data collected from the pilot study will be analysed to determine the internal consistency of the questionnaire items using Cronbach's Alpha coefficient. This statistical measure evaluates how closely related a set of items are as a group, thereby indicating the reliability of the instrument. The Cronbach's Alpha values range between 0 and 1, where higher values indicate greater reliability.

In this study, a Cronbach's Alpha coefficient of 0.70 or higher will be considered acceptable, as recommended in social science research. If the reliability coefficient is found to be below this threshold, necessary revisions will be made to the instruments by removing or modifying inconsistent items before conducting the main study.

Additionally, standard procedures will be followed during data collection to maintain consistency, including using the same instruments across all selected hospitals, providing clear instructions to respondents, and ensuring that data collection is conducted under similar conditions.

By ensuring reliability through pilot testing and statistical validation, the study will produce consistent and dependable data, thereby enhancing the overall quality and credibility of the research findings.

3.6 Data Analysis Methods

3.6.1 Data Analysis Methods

For data analysis, both quantitative and qualitative data analysis techniques will be employed to ensure a comprehensive understanding of maternal emergency response systems and the effectiveness of the proposed GPS-based solution.

Quantitative data collected through structured questionnaires will be coded and entered into Statistical Package for Social Sciences (SPSS) version 26 for analysis. Descriptive statistical methods will be used to summarize and interpret the data. These will include frequencies and percentages to describe the distribution of responses related to variables such as availability of emergency tools, system efficiency, and access to maternal health services. In addition, measures of central tendency, particularly mean values, will be used to analyse response times and assess the average efficiency of emergency response systems across the selected hospitals.

Further analysis will involve cross-tabulation to examine relationships between variables, such as the relationship between type of hospital (public or private) and response efficiency. This will help in identifying patterns and differences across facilities, thereby supporting the study objectives related to assessing current systems and identifying gaps.

Qualitative data obtained from interviews and observations will be analysed using thematic analysis. This process will involve transcribing interview responses, organizing the data into categories, and identifying recurring themes related to emergency response challenges, system limitations, and coordination gaps. Thematic analysis is a foundational method for qualitative research, providing the flexibility to interpret complex human-computer interactions in healthcare settings (Braun & Clarke, 2006).

The results of the data analysis will be presented using a combination of tables, charts, and narrative descriptions. Tables and charts will be used to present quantitative findings in a clear and organized manner, while narrative explanations will be used to interpret both quantitative and qualitative results. This approach will enhance clarity and ensures that findings are easily understood and directly linked to the research objectives.

Overall, the synergy of these analytical techniques will enable a robust assessment of current limitations while providing a data-driven foundation for the design, deployment, and performance monitoring of the GPS-based maternal system.

3.6.2 Data Quality Control

Maintaining high data quality is essential in this study as it will ensure that findings on maternal emergency response systems and the proposed GPS-based solution are valid and dependable.

Several measures will be implemented to ensure data quality during the study.

Daily data checks will be conducted during the data collection process. We will review completed questionnaires and interview records at the end of each day to ensure completeness, accuracy, and consistency. Any errors, omissions, or inconsistencies identified will be corrected immediately while still in the field to avoid data loss.

In addition, close supervision will be maintained throughout the data collection process. We, researchers will monitor how data is collected to ensure adherence to the study procedures and guidelines. This will include verifying that respondents are appropriately selected and that instruments are administered correctly.

Furthermore, standardized data collection procedures will be followed across all hospitals. The same instruments will be used consistently, and clear instructions will be provided to all respondents to minimize variation in responses.

These data quality control measures will help to minimize errors, reduce bias, and ensure that the data collected is accurate and reliable, thereby enhancing the overall credibility of the study findings.

3.8 Systems Study and Analysis Methods

A systems study will be conducted to map the existing maternal emergency response and referral workflows within Mukono General Hospital and surrounding health facilities. The research team will observe the current manual processes, trace actual patient referral flows, and interview emergency response coordinators.

To identify critical bottlenecks, a Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis will be executed. The current information flow—where spatial invisibility leads to delayed responses—will be modeled using context diagrams and manual process flowcharts. This systems study will form the clinical and logistical baseline for designing the new GPS-integrated interface.

3.9 System Requirements and Specification Methods

System requirements will be gathered using the findings from the stakeholders' surveys and interviews. Elicited requirements will be classified into User Requirements, Functional Requirements, Non-Functional Requirements, and System Specifications:

- **User Requirements:** Expectant mothers must be able to trigger an emergency alert with a single button press or a basic USSD code. Health workers must receive real-time notifications with map coordinates. Ambulance drivers must see optimized routing.
- **Functional Requirements:** The system will capture the expectant mother's real-time GPS coordinates, search for and notify the nearest ambulance or facility, securely store patient records in a SQL database, and display location data on a digitized map.
- **Non-Functional Requirements:** The system must maintain 99.9% uptime, locate patients within 5–10 meters accuracy, display simple and local language interfaces, and execute alert dispatches within 30 seconds.
- **System Requirements:** Server (minimum Intel i5, 8GB RAM, 500GB SSD), Client (Android-enabled mobile phone with GPS capability), Database (MySQL/PostgreSQL with PostGIS extension), Backend Languages (PHP/NodeJS), and Mobile App (Java/Kotlin).

3.10 Systems Design and Modeling Methods

The system design will translate user and functional requirements into structural models. The structural design of the GPS-Enabled Maternal Health System will be modeled using the following design methodologies:

- Entity-Relationship Diagrams (ERDs): To map the database tables, attributes, primary/foreign keys, and data relationships (Mothers, Medical Records, Hospitals, Responders, Emergency Logs).
- Data Flow Diagrams (DFDs): Context Level, Level 0, and Level 1 DFDs will be constructed to visualize the processes, data stores, data flows, and external entities of the tracking system.
- User Interface (UI) Wireframing: Sketching and layout designs of the mobile application emergency screen and the web-based hospital dashboard to ensure usability for semi-literate mothers.

3.11 System Implementation, Testing & Validation Methods

The implementation phase will involve developing a functional prototype based on the design specifications. The prototype development will use a modular coding approach, separating the mobile app module, database module, and hospital dashboard API module. Testing and validation will be executed in three stages:

- Unit Testing: Individually testing modules (GPS coordinate extraction, SMS dispatch engine, mapping API) to verify correct syntax and outputs.
- Integration Testing: Testing the combined modules to ensure seamless communication between the mobile app, backend server, and web dashboard database.
- System Validation: Testing the system through simulated maternal emergencies in Mukono District. Consenting VHTs and midwives will conduct mock emergency triggers from rural Nama and island Koome areas to measure network transmission time, location accuracy, and routing efficiency.

3.12 Ethical Considerations

Ethical issues in this study will be carefully addressed to ensure that data collection in hospitals within Mukono District is conducted in a responsible and professional manner.

Prior to data collection, ethical approval will be obtained from relevant authorities such as the Mukono District Health Office, and the management of the selected hospitals, including

Mukono Church of Uganda Hospital, C-Care (IMC) Hospital, AAR Pearl Hospital, and Mukono General Hospital. Obtaining these approvals will ensure that the study complies with national research regulations and institutional requirements.

Informed consent will be obtained from all participants before their involvement in the study. Participants will be clearly informed about the purpose of the research, the procedures involved, and their role in the study. They will also be made aware of their right to decline participation or withdraw from the study at any time without any negative consequences.

Confidentiality will be strictly maintained throughout the research process. Personal information collected from respondents will not be disclosed to unauthorized individuals, and responses will be anonymized to protect participants' identities. Data will be securely stored and used solely for academic purposes.

Participation in the study will be entirely voluntary. No participant will be forced or coerced into taking part in the research. The researcher will ensure that respondents feel comfortable and free to provide honest responses without fear of intimidation or bias.

Additionally, we will ensure that the study does not cause harm to participants. Data collection will be conducted in a respectful manner, particularly when dealing with expectant mothers and healthcare workers in sensitive hospital environments.

By adhering to these ethical principles, the study will uphold the highest standards of research integrity and ensure the protection of all participants involved.

3.12.1 Research Team Roles

- Nabukeera Zaharah Ramulah – Team Leader & Literature Review
- Muhinda Praise Francis– Field Coordinator & Data Collection
- Kato Hassan – Data Analyst & Report Writing

This division ensures efficiency, specialization, and timely completion of the study.

3.13 Chapter Summary

This chapter has detailed the research methodology employed in the study, encompassing the research design, study population, and sampling techniques. It described the mixed-methods approach, combining a descriptive cross-sectional design with a Prototyping System Development Model to ensure the technical solution is both evidence-based and user-centred. Furthermore, the chapter outlined the data collection instruments including questionnaires, interviews, and system testing, and the quality control measures used to maintain the validity and reliability of the findings. These methods provide a rigorous framework for evaluating the existing maternal emergency workflows and for the subsequent design, implementation, and testing of the GPS-based monitoring system in Mukono District

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APPENDICES

Appendix I: Research Questionnaire

RESEARCH TITLE: GPS-BASED MATERNAL EMERGENCY RESPONSE AND MONITORING SYSTEM FOR IMPROVING HEALTH IN UGANDA

Dear Respondent,

We are undergraduate students from Kyambogo University pursuing a Bachelor's Degree in Information Technology and Computing. As part of our academic requirements, we are conducting a research project to design and develop a GPS-based maternal emergency response and monitoring system for Mukono District. This system aims to reduce the delays maternal health services face when locating mothers in obstetric distress.

You have been selected to participate in this study. Your responses will remain strictly confidential and will be used solely for academic research purposes. Your participation is completely voluntary, and you are free to skip questions or withdraw at any time. We appreciate your time and contributions to this critical study.

SECTION A: DEMOGRAPHIC / PROFILE DATA

1. Category of Respondent:

- ☐ Healthcare Provider (Midwife / Nurse / Doctor)
- ☐ Emergency Responder (Ambulance Driver / VHT)
- ☐ Health Administrator / DHO staff
- ☐ Expectant or Postnatal Mother

2. Gender:

- ☐ Male
- ☐ Female

3. Age Group:

- ☐ Below 20 years
- ☐ 20–30 years
- ☐ 31–40 years
- ☐ 41–50 years
- ☐ Above 50 years

4. Primary Sub-county of Operations / Residence in Mukono District:

- ☐ Nama sub-county
- ☐ Koome Islands
- ☐ Mukono Municipality / Central
- ☐ Other (Specify: _____)

SECTION B: ASSESSMENT OF CURRENT EMERGENCY WORKFLOWS (Likert Scale)

For each of the following statements, please check the box that best describes your opinion. (Scale: 1 = Strongly Disagree [SD], 2 = Disagree [D], 3 = Neutral [N], 4 = Agree [A], 5 = Strongly Agree [SA])

1. Finding the physical location of expectant mothers in distress during an emergency is a major challenge due to lack of addresses.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

2. Vague verbal directions and unmapped rural roads frequently delay ambulance arrivals.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

3. Lack of spatial location visibility causes delays that put expectant mothers at risk of life-threatening complications (for example, PPH).

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

4. Referral loops (bouncing mothers between health facilities) are common because we lack real-time resource and location coordination.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

5. The current communication method (manual records, standard calls) is inadequate for rapid maternal emergency dispatch.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

SECTION C: REQUIREMENTS FOR THE PROPOSED GPS SYSTEM

1. How do you currently locate a patient during an emergency? (Check all that apply)

- ☐ Verbal descriptions and landmarks
- ☐ Calling local Village Health Teams (VHTs) for guidance
- ☐ Patient registration records / paper cards
- ☐ WhatsApp / Google Maps location sharing (where possible)

- ☐ No structured method, we wait for them to arrive
2. What are the key features you would expect in a maternal tracking system?
(Check all that apply)
- ☐ One-touch emergency alert button for mothers (USSD / SMS capability)
 - ☐ Real-time GPS tracking displaying mother's location on hospital map
 - ☐ Central dashboard showing ambulance and VHT proximity to patient
 - ☐ Direct communication / chat option between mother, responder, and clinic
 - ☐ Patient's basic clinical history transmitted automatically during alert
3. What is the biggest barrier you foresee in adopting a GPS-based system in Mukono? (Select one)
- ☐ Poor mobile network and internet connectivity (especially in island areas)
 - ☐ Low smartphone ownership and digital illiteracy among expectant mothers
 - ☐ High cost of system implementation and smartphone maintenance
 - ☐ Resistance to technology adoption by healthcare staff / VHTs
 - ☐ Concerns over patient location data privacy and security

SECTION D: PERCEIVED TECH-ACCEPTANCE (TAM METRICS)

Please rate your agreement with the following statements regarding the potential implementation of the GPS Maternal Tracking System. (Scale: 1 = SD, 2 = D, 3 = N, 4 = A, 5 = SA)

1. Using a GPS maternal tracking system would make emergency dispatch significantly faster and reduce travel delays.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

2. The system would allow healthcare workers to prepare the necessary clinical resources before the patient arrives.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

3. An intuitive mobile application with minimal steps would be easy for expectant mothers to use during a crisis.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

4. A dashboard displaying Google Maps routes would be easy for ambulance drivers and VHTs to navigate.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

5. I would strongly support the implementation of this GPS tracking system in Mukono District.

☐ 1 (SD) ☐ 2 (D) ☐ 3 (N) ☐ 4 (A) ☐ 5 (SA)

Appendix II: Estimated Budget

The table below shows the estimated costs associated with conducting this study and developing the GPS Maternal Emergency Response and Monitoring System prototype. The total cost is estimated at UGX 3,250,000.

Item Description	Quantity	Unit Cost (UGX)	Total Cost (UGX)
Software Development Tools (Server hosting, API licenses, mapping SDKs)	Lump Sum	N/A	800,000
Field Surveys & Data Collection (Travel & subsistence in Nama & Koome sub-counties)	3 Researchers	250,000	750,000
Internet Data & Airtime (For software testing and field coordinator communications)	3 months	100,000	300,000
User Validation Testing (Mock emergency runs with ambulance drivers & VHTs)	5 runs	120,000	600,000

Stationery, Printing, and Book Binding (Drafts, guidelines, final project reports)	6 copies	50,000	300,000
Contingency Funds (Unforeseen expenses, minor hardware adjustments)	Lump Sum	N/A	500,000
GRAND TOTAL			3,250,000

Thank you for your valuable feedback.